UCLouvain, CP3
Louvain-la-Neuve, Belgium
soumen.roy@uclouvain.be
soumenroy.github.io

Soumen Roy

Research Profile

I am a postdoctoral researcher at UCLouvain, working on gravitational-wave astronomy and an active member of the LIGO–Virgo–KAGRA Collaboration. My research focuses on tests of general relativity, astrophysical environments surrounding compact binaries, waveform modeling, and Bayesian inference. I develop data-analysis frameworks, algorithms, and statistical methods to probe subtle physical effects in gravitational-wave observations with current and future observatories.

Academic Appointments

2024–Present **Postdoctoral Researcher**, Centre for Cosmology, Particle Physics and Phenomenology (CP3), UCLouvain, Belgium.

2020–2024 **Postdoctoral Researcher**, Nikhef / Utrecht University, The Netherlands.

Education

2014–2021 **Ph.D.**, *Indian Institute of Technology Gandhinagar*, India

- *Thesis title*: Aspects of gravitational wave data analysis from compact binary coalescences
- *Supervisor*: Prof. Anand S. Sengupta

2011–2013 **M.Sc. in Physics**, *The University of Calcutta*, India

- *Specialization*: Astrophysics & Astroparticle Physics
- *Thesis title*: An Introduction to Fluid Dynamics and Numerical Solution of Shock Tube Problem by Using Roe Solver
- *Supervisor*: Prof. Sanjay Kumar Ghosh

2008–2011 **B.Sc. (Honours)**, *Vidyasagar Unversity*, India,
Specialization: Physics

Research Program

- Advance precision tests of gravity with gravitational waves by developing inference frameworks that incorporate higher-order multipoles, precession, and parametrized deviations from general relativity while controlling waveform systematics.
- Probe compact binaries beyond the standard vacuum paradigm through waveform and inference models for dense gas, ultralight scalar-field environments, and line-of-sight acceleration, with the goal of distinguishing genuine environmental signatures from modelling bias.
- Build analysis tools for subtle or unexpected signal features, including semi-model-dependent wavelet reconstruction and fast parameter-estimation methods for complex compact-binary sources.
- Improve search infrastructure for current detectors through template-bank methodology and related compact-binary analysis pipelines.
- Prepare for next-generation gravitational-wave science through work on Einstein Telescope analysis tools, eccentric-binary inference, and exploratory detection concepts in the μHz band using GNSS constellations.

Computational Expertise

Programming and workflow	Python, C/C++, HTCondor, bash, LaTeX; experience with MATLAB, Mathematica, Gnuplot, and FORTRAN.
Gravitational-wave software	LALSuite, PyCBC, Bilby, BayesWave; Bayesian inference and waveform-model implementation.
Computing environments	Linux, macOS, Windows; high-throughput and high-performance computing for large-scale gravitational-wave analyses.

Mentoring

Graduate students:

2025–Current Jef Heynen (*CP3, UCLouvain*) with Justin Janquart
Leveraging the null stream to detect strongly lensed gravitational waves signals

2025–Current Ioannis Liodis (*Deutsches Elektronen-Synchrotron DESY*) with Samaya Nissanke
Impact of astrophysical environment surrounding compact binaries on cosmology inference

2022–2024 Giada Caneva Santoro (*IFAE in UAB*) with Mario Martinez
Measuring the astrophysical environment surrounding the compact binaries with second generation and future detectors

2022–2025 Abhishek Sharma (*IIT Gandhinagar*) with Anand S. Sengupta,
Designing a search framework for detecting signals beyond general relativity, developing rapid parameter estimation framework

2021–2022 Anna Pucher (*Nikhef / Utrecht University*) with Chris Van Den Broeck,
Testing general relativity using higher-order modes of gravitational waves from binary black holes

Undergraduate and Master's students:

2025 Ioannis Liodis (*GRAPPA, University of Amsterdam*) with Samaya Nissanke
Impact of astrophysical environment surrounding compact binaries on cosmology inference

2022–2023 Bram van de Zande (*Utrecht University*) with Chris Van Den Broeck,
Black holes, or black hole mimickers (superspinar)? Gravitational waves as a means to tell the difference.

2021 Marien Hofstee (*Utrecht University*) with Chris Van Den Broeck & Fabian Ziltener,
Higher harmonics in gravitational wave signals from binary black hole mergers

2019–2021 Harsh Narola (*IIT Gandhinagar*) with Anand S. Sengupta,
Beyond general relativity: Designing a template-based search for exotic gravitational wave signals

2019 Mohini Umesh Ramwala (*IIT Gandhinagar*) with Anand S. Sengupta,
GR and Non GR template banks for LIGO data analysis

Teaching skills

2017 Spring *Graduate Teaching Fellow at IIT Gandhinagar*: Statistical Mechanics

2018 Fall *Graduate Teaching Fellow at IIT Gandhinagar*: Computational Physics

2019 Spring *Graduate Teaching Fellow at IIT Gandhinagar*: Introduction to computing

Selected Talks and Presentations

16. *Line-of-sight acceleration in compact binaries: the role of higher harmonics and eccentricity*,
XVI Einstein Telescope Symposium, Achen, Germany, *June, 2026*

16. *Scalar Fields Around Black Hole Binaries in LIGO-Virgo-KAGRA*,
New opportunities to detect gravitational waves and dark matter in orbital dynamics, Benasque Science Center, Spain *December, 2025*

15. *Fast likelihood evaluation of eccentric–precessing binaries using relative binning*,
4th Einstein Telescope Annual Meeting, Opatija, Croatia *December, 2025*

14. *Testing modified gravity with the eccentric neutron star–black hole merger GW200105*,
Belgian-Dutch Gravitational Wave Meeting, Radboud University Nijmegen, Netherlands *December, 2025*

13. *Testing modified gravity with the eccentric neutron star–black hole merger GW200105*,
Joint Theoretical Gravitational Wave Seminars – June,
Leuven, Belgium *December, 2025*

12. *Parametrized test of general relativity with gravitational waves: including higher harmonics and precession*,
Belgian Physical Society general scientific meeting,
Louvain-La-Neuve, Belgium *December, 2025*

11. *Probing gravitational waves using GNSS constellations*,
Tabletop Scale Cosmology, online conference *December, 2025*

10. *Probing gravitational waves using GNSS constellations*,
59th Rencontres de Moriond on Gravitation: Moriond 2025 Gravitation,
La Thuile, Aosta Valley, Italy *December, 2025*

9. *Can Compact Binary Coalescences in Environments pose as in Vacuum?*,
biweekly BelGrav meeting, online meeting *December, 2024*

8. *Update on TIGER analysis with GW230814*,
LIGO-Virgo-KAGRA Collaboration Meeting, 2024,
Barcelona, Spain *September, 2024*

7. *A non-orthogonal wavelet transformation for reconstructing gravitational wave signals*,
APS April Meeting, NY. (April 2022) *April 10, 2022*

6. *Detection of non-quadrupole modes of gravitational waves from multiple inspiralling blackholes*,
9th International Conference on Gravitation and Cosmology,
IISER Mohali, India *December 11, 2019*

5. *Effectual template banks for upcoming compact binary searches in Advanced-LIGO Virgo data*,
30th meeting of the Indian Association for General Relativity and Gravitation,
BITS Pilani, Hyderabad, India, *January 3, 2019*

4. *Designing an effectual template placement algorithm for searches of gravitational wave from compact binary coalescences*,
Exploring the Universe: Near Earth Space Science to Extra-Galactic Astronomy,
SNBNCBS Kolkata, India, *November 16, 2018*

3. *Designing an effectual template placement algorithm for gravitational wave searches*,
Summer School: Gravitational Waves 2018,
Ecole de Physique des Houches, F-74310 LES HOUCHES, France, *July 25, 2018*

2. *Hybrid geometric-random template placement algorithm for gravitational wave searches from compact binary coalescences*
29th meeting of the Indian Association for General Relativity and Gravitation,
IIT Guwahati, India, *18th May 2017*

1. *A hybrid geometric-random template placement algorithm for gravitational wave searches from compact binary coalescences*,
Time Series Analysis for Synoptic Surveys and Gravitational Wave Astronomy,
ICTS, Bangalore, India *March 23, 2017*

Professional memberships

Since 2020 Member of Einstein Telescope Observational Science Board

Since 2020 Member of Virgo Scientific Collaboration

2017–2020 Member of LIGO-India Scientific Collaboration

Since 2017 Member of LIGO Scientific Collaboration

List of publications

∗∗ Full list of publications is available on InspireHEP, and short author list publications can also be found on InspireHEP

Short author-list publications

24. **Soumen Roy** and Justin Janquart,
Line-of-sight acceleration in compact binaries with higher harmonics and eccentricity,
arXiv:2606.08838, *Submitted to Phys. Rev. D*

23. **Soumen Roy** *et al.*,
Scalar fields around black hole binaries in LIGO-Virgo-KAGRA,
Phys. Rev. Lett. **136**, 191402, (2026), arXiv:2510.17967

22. Jef Heynen, **Soumen Roy**, and Justin Janquart,
Leveraging the null stream to detect strongly lensed gravitational waves signals,
J. Phys. Conf. Ser. **3177** (2026) 1, 012080, arXiv:2509.06745

21. Parthapratim Mahapatra *et al.* inc. **Soumen Roy**,
Confronting general relativity with principal component analysis: Simulations and results from GWTC-3 events,
Phys. Rev. D **112**, 104007 (2025), arXiv:2508.06862

20. Abhishek Sharma, Lalit Pathak, **Soumen Roy**, and Anand S. Sengupta,
Rapid parameter estimation with the full symphony of compact binary mergers using meshfree approximation,
Accepted in Phys. Rev. D 12 June,(2026), arXiv:2508.04172

19. Debtroy Das, **Soumen Roy**, *et al.*,
Probing missing physics from inspiralling compact binaries via time-frequency tracks,
Eur. Phys. J. C **86**, 667 (2026) arXiv:2507.21566

18. **Soumen Roy** and Justin Janquart,
Testing modified gravity with the eccentric neutron star-black hole merger GW200105,
Phys. Rev. D **113**, 024056 (2026), arXiv:2507.21315

17. **Soumen Roy**, Bruno Bertrand, and Justin Janquart,
Probing gravitational waves using GNSS constellations,
59th Rencontres de Moriond on Gravitation:Moriond 2025 Gravitation, 2025, arXiv:2505.21716

16. **Soumen Roy** *et al.*,
Improved parametrized test of general relativity using the IMRPhenomX waveform family: Including higher harmonics and precession,
Phys. Rev. D **113**, 024016 (2026), arXiv:2504.21147

15. **Soumen Roy** and Rodrigo Vicente,
Compact binary coalescences in dense gaseous environments can pose as ones in vacuum,
Phys. Rev. D **111**, 084037 (2025), arXiv:2410.16388

14. Justin Janquart *et al.* inc. **Soumen Roy**,
What is the nature of GW230529? An exploration of the gravitational lensing hypothesis,
Monthly Notices of the Royal Astronomical Society **537**, 1001-1014 (2024), arXiv:2409.07298

13. Elise Sänger, **Soumen Roy**, et al., *Tests of General Relativity with GW230529: a neutron star merging with a lower mass-gap compact object*,
Phys. Rev. D 113, 084070 (2026), arXiv:2406.03568

12. Giada Caneva Santoro, **Soumen Roy**, Rodrigo Vicente, *et al.*,
First constraints on binary black hole environments with LIGO-Virgo observations,
PoS EPS-HEP2023 (2024) 068

11. Abhishek Sharma, **Soumen Roy**, and Anand S. Sengupta,
Template bank to search for exotic gravitational wave signals from astrophysical compact binaries,
Phys. Rev. D **109**, 124049 (2024), arXiv:2311.03274

10. Giada Caneva Santoro, **Soumen Roy**, Rodrigo Vicente, *et al.*,
First constraints on compact binary environments from LIGO-Virgo data,
Phys. Rev. Lett. **132**, 251401 (2024) , arXiv:2309.05061

9. Sushant Sharma Chaudhary *et al.* (including **Soumen Roy**),
Low-latency gravitational wave alert products and their performance in anticipation of the fourth LIGO-Virgo-KAGRA observing run,
Proc. Nat. Acad. Sci. **121**, e2316474121 (2024) arXiv:2308.04545

8. Anna Puecher, Tim Dietrich, Ka Wa Tsang, Chinmay Kalaghatgi, **Soumen Roy**, *et al.*,
Unraveling information about supranuclear-dense matter from the complete binary neutron star coalescence process using future gravitational-wave detector networks,
Phys. Rev. D **107**, 124009 (2023), arXiv:2210.09259

7. Anna Puecher, Chinmay Kalaghatgi, **Soumen Roy**, *et al.*
Testing general relativity using higher-order modes of gravitational waves from binary black holes,
Phys. Rev. D **106**, 082003 (2022), arXiv:2205.09062

6. Harsh Narola, **Soumen Roy**, and Anand S. Sengupta,
Beyond general relativity: Designing a template-based search for exotic gravitational wave signals,
Phys. Rev. D **107**, 024017 (2023), arXiv:2207.10410

5. **Soumen Roy**,
Nonorthogonal wavelet transformation for reconstructing gravitational wave signals
Phys. Rev. Research **4**, 033078 (2022), arXiv:2201.01526

4. **Soumen Roy**, Anand S. Sengupta, and K. G. Arun,
Unveiling the spectrum of inspiralling binary black holes,
Phys. Rev. D **103**, 064012 (2021), arXiv:1910.04565.

3. **Soumen Roy**, Anand S. Sengupta, and Parameswaran Ajith,
Effectual template banks for upcoming compact binary searches in Advanced-LIGO and Virgo data,
Phys. Rev. D **99**, 024048 (2019), arXiv:1711.08743.

2. **Soumen Roy**, Anand S. Sengupta, and Parameswaran Ajith,
Geometric-stochastic template banks for gravitational wave searches from compact binaries in advanced-LIGO data, 42nd COSPAR Scientific Assembly, Vol. **42** (2018) pp. E1.15-36-18.

1. **Soumen Roy**, Anand S. Sengupta, and Nilay Thakor, *Hybrid geometric-random template-placement algorithm for gravitational wave searches from compact binary coalescences*,
Phys. Rev. D **95**, 104045 (2017), arXiv:1702.06771.

Collaboration papers

I am also an author of many LIGO–Virgo–KAGRA (LVK) collaboration publications and analyses. Major responsibilities include parametrized tests of general relativity, higher-order-mode analyses, compact-binary search template banks, and exceptional-event studies. Below, I briefly describe my contributions and list the papers for which I contributed significantly to either the analysis or the writing.

– Lead developer and primary analyst for TIGER-based parametrized tests of general relativity; reviewer for principal-component-analysis tests; developer and reviewer for higher-order-mode tests; contributor to template-bank construction for online and offline compact-binary searches; and member of the liaison/paper-writing team for exceptional-event analyses.
– Contributed extensively to template-bank construction for compact-binary searches with PyCBC-Live, PyCBC-Offline, and MBTA since the third LIGO–Virgo observing run. These template banks are generated using our hybrid geometric-random placement method. In the fourth observing run, this work expanded to include more intermediate-mass black hole systems in PyCBC-Offline searches, improving search sensitivity.
– Played a leading role in the first detection of higher harmonics in GW190412. Our time-frequency method provided the first evidence without relying on parameter-estimation analyses. The same framework later found strong evidence of higher harmonics in GW190814. Led method development, code implementation, and the LIGO–Virgo internal review process.

List of LIGO-Virgo-KAGRA collaboration publications

15. LIGO Scientific Collaboration, VIRGO Collaboration, and KAGRA Collaboration (including **Soumen Roy**),
GW240925 and GW250207: Astrophysical calibration of gravitational wave detectors,
Accepted in Phys. Rev. Lett. 30 April, 2026, arXiv:2605.11703

14. LIGO Scientific Collaboration, VIRGO Collaboration, and KAGRA Collaboration (including **Soumen Roy**),
GW230814: investigation of a loud gravitational-wave signal observed with a single detector,
Astrophys. J. Lett. **1004** (2026) 2, L23, arXiv:2509.07348

13. LIGO Scientific Collaboration, VIRGO Collaboration, and KAGRA Collaboration (including **Soumen Roy**),
Black Hole Spectroscopy and Tests of General Relativity with GW250114,
Phys. Rev. Lett. **136**, 041403 (2026), arXiv:2509.08099

12. LIGO Scientific Collaboration, VIRGO Collaboration, and KAGRA Collaboration (including **Soumen Roy**),
GWTC-4.0: Updating the Gravitational-Wave Transient Catalog with Observations from the First Part of the Fourth LIGO-Virgo-KAGRA Observing Run,
Astrophys. J. Lett. **1004** (2026) 2, L22, arXiv:2508.18082

11. LIGO Scientific Collaboration, VIRGO Collaboration, and KAGRA Collaboration (including **Soumen Roy**),
GWTC-4.0: Methods for Identifying and Characterizing Gravitational-wave Transients,
Astrophys. J. Lett. **1004** (2026) 2, L21, arXiv:2508.18081

10. LIGO Scientific Collaboration, VIRGO Collaboration, and KAGRA Collaboration (including **Soumen Roy**),
GWTC-4.0: An Introduction to Version 4.0 of the Gravitational-Wave Transient Catalog,
Astrophys. J. Lett. **995**, L18 (2025), arXiv:2508.18080

9. LIGO Scientific Collaboration, VIRGO Collaboration, and KAGRA Collaboration (including **Soumen Roy**),
Observation of Gravitational Waves from the Coalescence of a $2.5-4.5M_\odot$ Compact Object and a Neutron Star,
Astrophys. J. Lett. **970**, L34 (2024) arXiv:2404.04248

8. LIGO Scientific Collaboration and Virgo Collaboration (including **Soumen Roy**),
GWTC-3: Compact Binary Coalescences Observed by LIGO and Virgo During the Second Part of the Third Observing Run,
Phys. Rev. X **13**, 041039 (2023) arXiv:2111.03606

7. LIGO Scientific Collaboration and Virgo Collaboration (including **Soumen Roy**),
GWTC-2.1: Deep Extended Catalog of Compact Binary Coalescences Observed by LIGO and Virgo During the First Half of the Third Observing Run,
Phys. Rev. D **109**, 022001 (2024) arXiv:2108.01045

6. LIGO Scientific Collaboration and Virgo Collaboration (including **Soumen Roy**),
Observation of gravitational waves from two neutron star-black hole coalescences,
Astrophys. J. Lett. **915** L5 (2021) arXiv:2106.15163

5. LIGO Scientific Collaboration and Virgo Collaboration (including **Soumen Roy**),
GWTC-2: Compact Binary Coalescences Observed by LIGO and Virgo During the First Half of the Third Observing Run,
Phys. Rev. X **11**, 021053 (2021), arXiv:2010.14527,

4. LIGO Scientific Collaboration and Virgo Collaboration (including **Soumen Roy**),
GW190521: A Binary Black Hole Merger with a Total Mass of $150M_\odot$,
Phys. Rev. Lett. **125**, 101102 (2020), arXiv:2009.01075

3. LIGO Scientific Collaboration and Virgo Collaboration (including **Soumen Roy**),
GW190814: Gravitational Waves from the Coalescence of a $23M_\odot$ Black Hole with a $2.6M_\odot$ Compact Object,
Astrophys. J. Lett. **896**, L44 (2020), arXiv:2006.12611 (2020).

2. LIGO Scientific Collaboration and Virgo Collaboration (including **Soumen Roy**), *GW190412: Observation of a Binary-Black-Hole Coalescence with Asymmetric Masses* , Phys. Rev. D **102**, 043015 (2020), arXiv:2004.08342 (2020).
1. LIGO Scientific Collaboration and Virgo Collaboration (including **Soumen Roy**), *GW190425: Observation of a Compact Binary Coalescence with Total Mass* $\sim 3.4 M_{\odot}$, Astrophys. J. Lett. **892**, L3 (2020), arXiv:2001.01761 (2020).

Journal referee

- Physical Review D, Physical Review Letters